

Contest Math 8: Intro to Geometry (27079)

Connor Gordon

Sunday

Feb 16, 2025 - Nov 9, 2025

12:45 - 2:45 PM ET

Homework

Meeting: 1 2 3 4 **5** 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Homework: Meeting 5

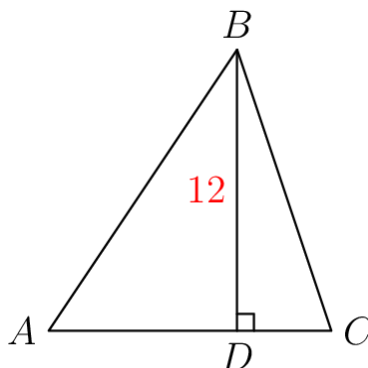
You have completed **9** of **15** challenge problems.
Due **Mar 23**.

Readings

Meeting 5: Chapter 4**Meeting 6:** Sections 5.1-5.4

Challenge Problems

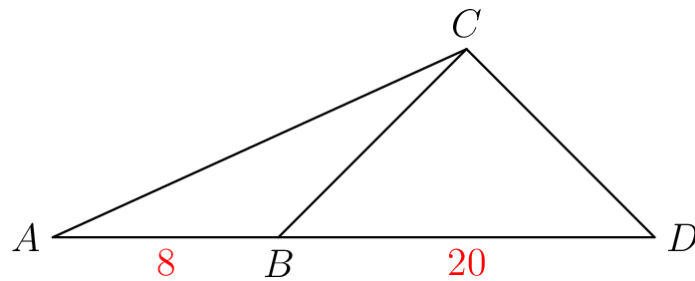
Problem 1 – Correct! – Score: 7 / 7 (28329)

Problem:Triangle ABC has area 16. Find length AC .[Report Error](#)*(Based on problem 4.9 in the textbook.)***Solution:**

Problem 2 – Correct! – Score: 7 / 7 (28330)

Problem:[Report Error](#)

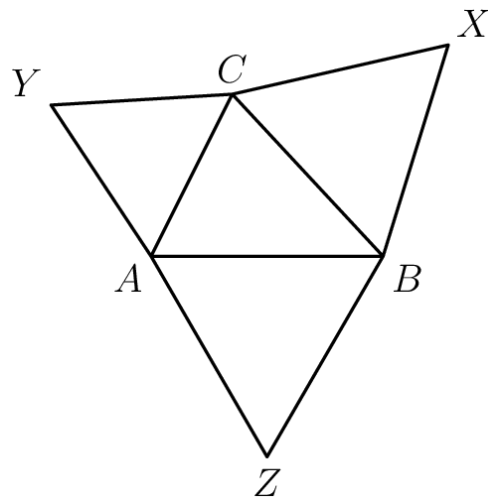
Given that $[ABC] = 10$, find $[BCD]$.



(Based on problem 4.10 in the textbook.)

Solution:**Problem:**[Report Error](#)

Let ABC be any triangle. Equilateral triangles BCX , ACY , and BAZ are constructed such that none of these triangles overlaps triangle ABC . The perimeter of hexagon $AZBXC Y$ is 60. What is the perimeter of triangle ABC ?



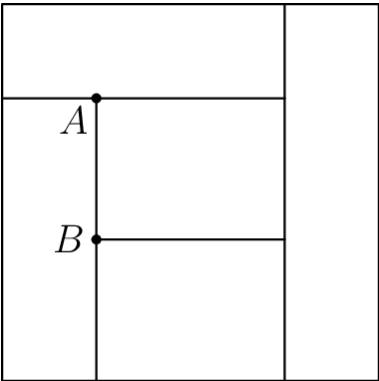
Solution:

.....

Problem:

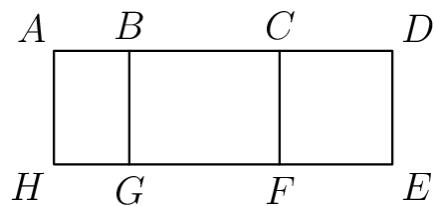
[Report Error](#)

A square is divided into five rectangles, and all the rectangles have the same area. Given that $AB = 1$, find the side length of the square.



Problem:[Report Error](#)

Rectangle $ADEH$ is divided into rectangles $ABGH$, $BCFG$, and $CDEF$. If $AC = 15$, $GE = 30$, $[ABGH] = 8$, and $[CDEF] = 28$, find $[BCFG]$.



F

Problem:[Report Error](#)

A rectangle is divided into four small rectangles as shown. Three of the small rectangles are labelled with their **perimeter**. What is the perimeter of the fourth rectangle?

10	?
6	25

Solution:

Problem:[Report Error](#)

A rectangle is divided into four small rectangles as shown. Three of the small rectangles are labelled with their **area**. What is the area of the fourth rectangle?

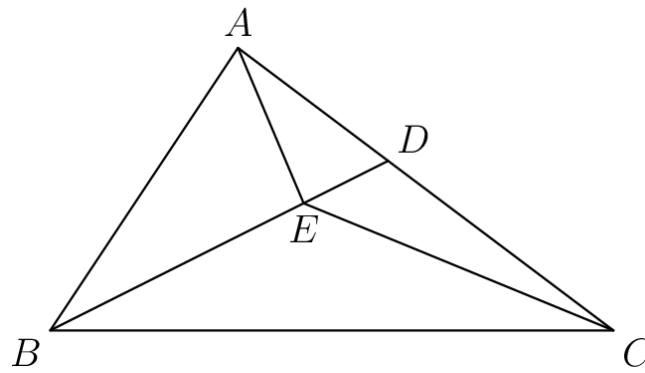
12	?
8	27

Solution:

Problem 8 – Correct! – Score: 7 / 7 (39008)

Problem:[Report Error](#)

In triangle ABC , $[ADE] = 4$, $[ABE] = 14$, and $[CDE] = 6$. Compute $[BCE]$.

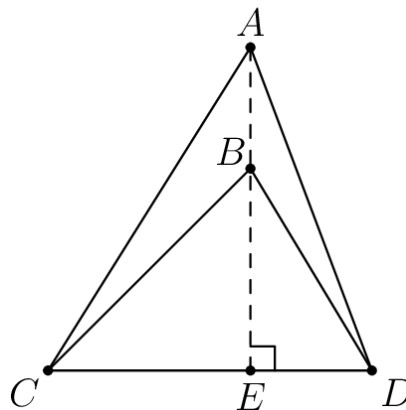


Problem 9 – Correct! – Score: 5 / 7 (10385)

Problem:

In the diagram, $BE = 7$, $AB = 6$, and $[CBDA] = 15$. What is $[CDA]$?

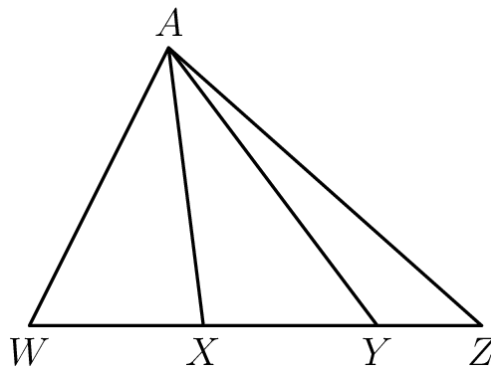
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Problem 10 (10389)

Problem:[Report Error](#)

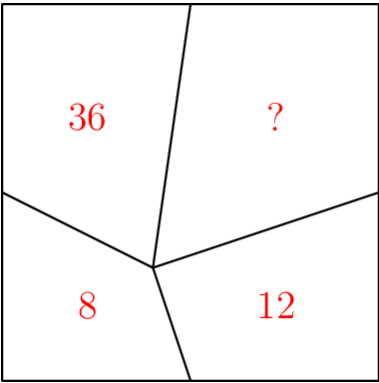
In the diagram, $WY = 12$, $XZ = 10$, $[AWX] = 15$, and $[AYZ] = 8$. Find $[AXY]$.



Problem 11 (39006)

Problem:[Report Error](#)

A point in a square is connected to the midpoints of its sides, dividing the square into four regions. The areas of three of the regions are labelled. Find the area of the fourth region.

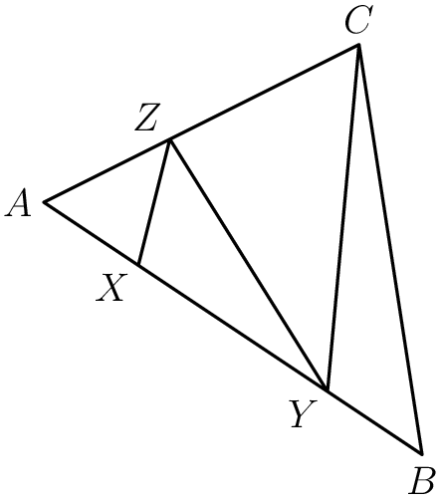


Your Answer

Problem 12 (10417)

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Problem:
In the diagram, $XY = 2 \cdot AX = 2 \cdot YB$ and $\frac{AZ}{ZC} = \frac{2}{5}$. Find $\frac{[YBC]}{[ZYC]}$.



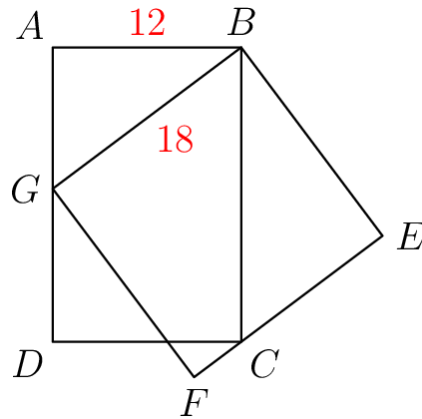
Your Answer

[Tips](#)

Problem 13 (39000)

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Problem:
In the diagram, $ABCD$ is a rectangle, and $BEFG$ is a square. Find AD .



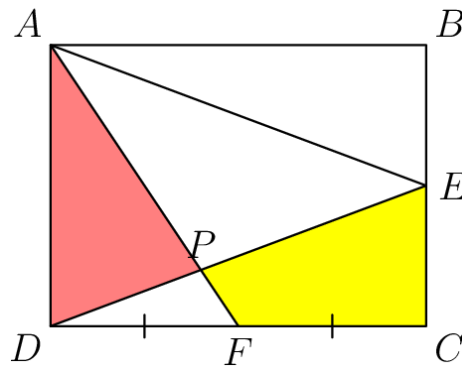
Your Answer

Problem 14 (39009)

Problem:

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In rectangle $ABCD$, $\frac{[AEP]}{[DFP]} = 5$. Find the ratio of the area of the red region to the area of the yellow region. Enter your answer as a fraction.



Your Answer

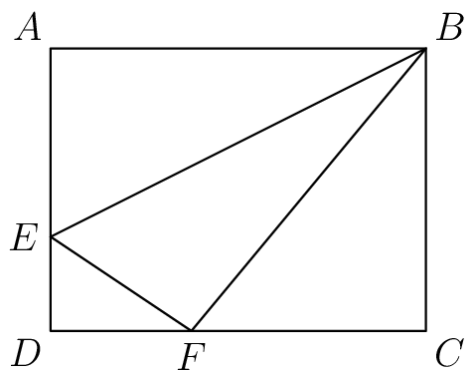
[Tips](#)

Problem 15 (39007)

Problem:

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In rectangle $ABCD$, $[ABE] = [BEF] = [BFC]$. Compute $\frac{AE}{ED}$.

**Beast Problem:** 🐾

This is a **Beast** problem. Answering this problem can only increase your score. You will not lose points for skipping this problem or answering incorrectly.

[Tips](#)

Book Problems (39027)

In addition to the homework listed here each week, you also need to complete several additional homework problems from your textbooks. This week's book problems are:

- *Introduction to Geometry*: 4.23, 4.24, 4.32, 4.38
- *Art of Problem Solving, Vol. 1*: Problem 265

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